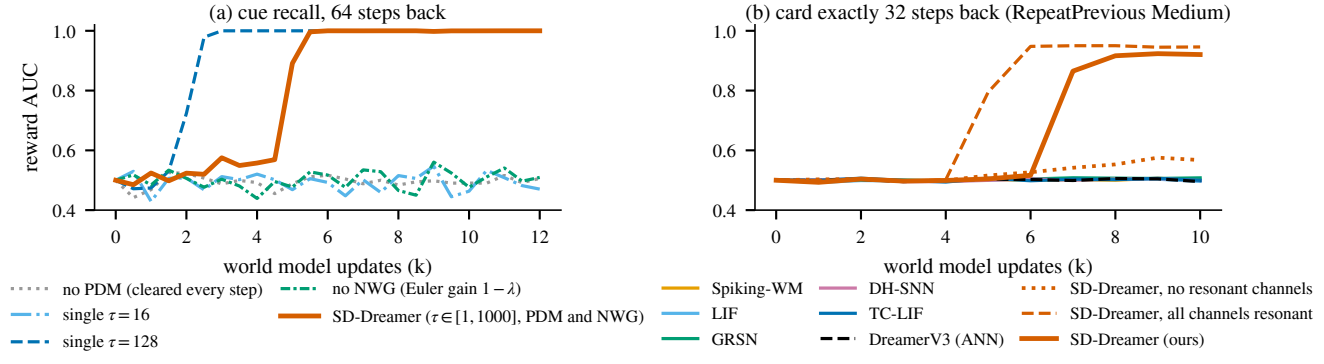
(a) cue recall, 64 steps back
reward AUC
1.0
0.8
0.6
0.4
0
2
4
6
8
10
12
world model updates (k)
(b) card exactly 32 steps back (RepeatPrevious Medium)
1.0
0.8
0.6
0.4
0
2
4
6
8
10
world model updates (k)
no PDM (cleared every step)
single $\tau = 16$
single $\tau = 128$
no NWG (Euler gain $1 - \lambda$)
SD-Dreamer ($\tau \in [1, 1000]$, PDM and NWG)
Spiking-WM
LIF
GRSN
DH-SNN
TC-LIF
DreamerV3 (ANN)
SD-Dreamer, no resonant channels
SD-Dreamer, all channels resonant
SD-Dreamer (ours)